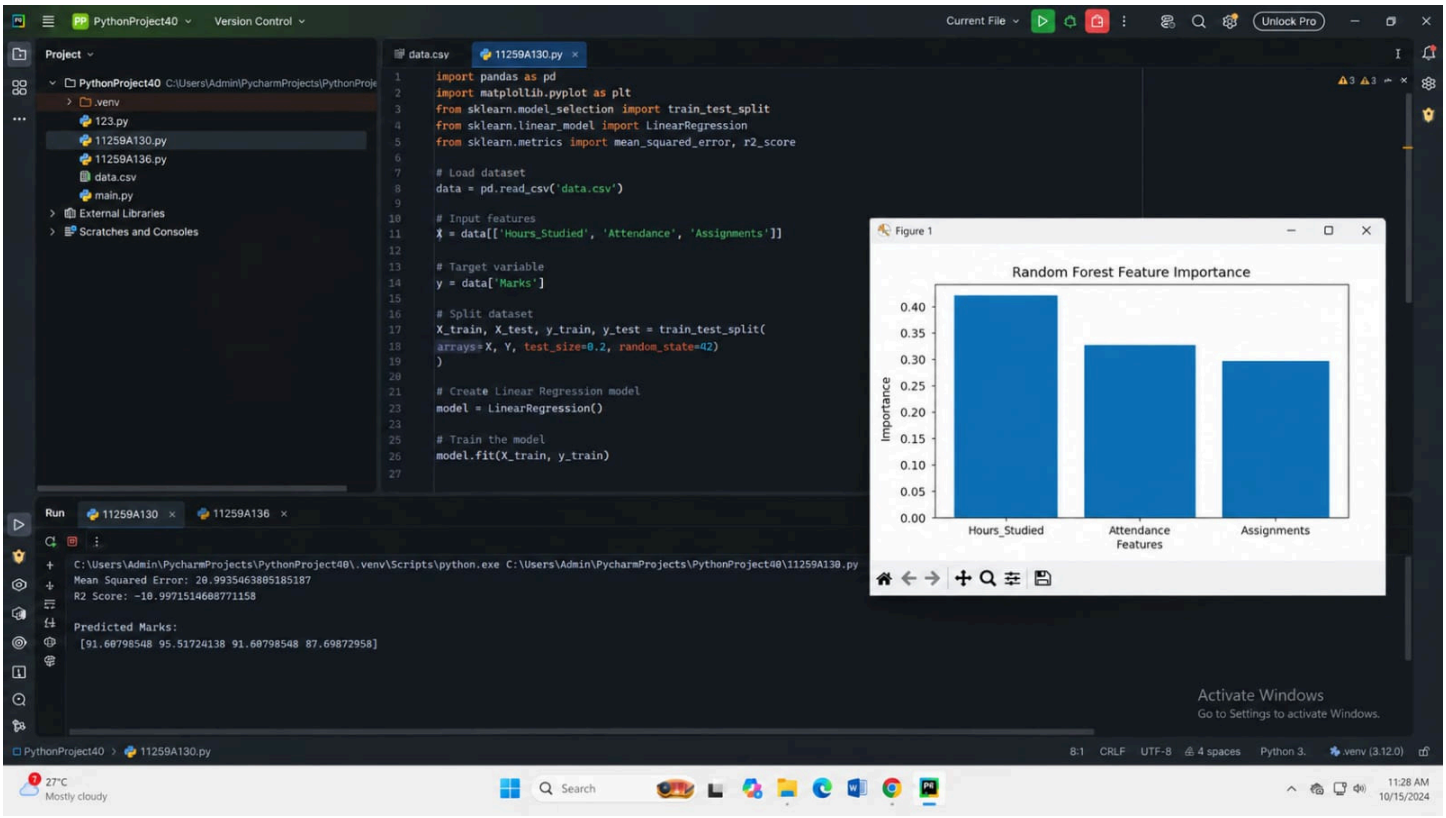
Experiment 7: Implement Random Forest and Analyze Performance

Aim

To implement a Random Forest classification algorithm to predict students' results based on Hours_Studied, Attendance, and Assignments, evaluate the model's performance, and visualize the importance of each feature.

Algorithm

1. Import the required libraries: Pandas, train_test_split, RandomForestClassifier, evaluation metrics, and Matplotlib.
2. Load the dataset from the CSV file data.csv.
3. Select the input features Hours_Studied, Attendance, and Assignments and store them in X.
4. Select the target variable Result and store it in y.
5. Split the dataset into training and testing sets using train_test_split(), with 80% for training and 20% for testing.
6. Create a Random Forest classifier using RandomForestClassifier(n_estimators=100, random_state=42).
7. Train the model using model.fit(X_train, y_train).
8. Predict the results for the testing dataset using model.predict(X_test).
9. Evaluate the model using accuracy_score(), confusion_matrix(), and classification_report().
10. Extract feature importance values using model.feature_importances_.
11. Plot the feature importance values using a bar chart.
12. Add axis labels and a title, and display the graph using plt.show().



Result

The Random Forest classification model was successfully trained and evaluated. Based on the provided output, the model achieved an **accuracy of 95%** on the test dataset.